OXFORD AQA
INTERNATIONAL QUALIFICATIONS

The international exam board *that puts fairness first*

International AS and A-level

Physics (9630) PH02
Unit 2 Electricity, Waves and Particles
Report on the Examination

June 2026

OxfordAQA is a partnership between AQA, the UK's largest awarding body, and Oxford University Press, a department of the University of Oxford. We offer globally relevant international GCSEs, AS and A-levels to schools across Europe, Africa and Asia.

REPORT ON THE EXAMINATION: INTERNATIONAL AS/A-LEVEL PHYSICS (9630) UNIT 2 ELECTRICITY, WAVES AND PARTICLES – JUNE 2026

This paper was accessible to the majority of students. The mean mark increased by 4 compared to Summer 2025 but was the same as January 26. There was a large standard deviation, showing that the cohort had a wide spread of marks.

As in most previous series, students performed slightly better on average in the multiple-choice section C than in Sections A and B. In every section C question, the most popular response was the correct one.

Students gave some well-worded explanations particularly in the electricity topic. Some misconceptions were seen in the optical fibres question.

Some general principles to note that apply in every series:

- In Section A of the paper, students may be penalised if they quote a numerical answer to just one significant figure, where data is given to more than one significant figures. An example in this paper was Q7.1. In Section B of the paper, in the data analysis question, further rules for significant figures apply as described in the practical handbook.
- If the question word is Determine, the student must show their calculation workings to obtain full marks. This applied, for example, in Q10.2 and 10.4 of this paper. In Q10.4, a bald answer with no working scored no marks.
- In an explanation question, students should refer to a relevant equation where possible to justify their answer. This applied to Q7.4 and Q8.3 in this paper.

QUESTION A1

Around two thirds of students scored the mark for this base unit question. A common error was quoting the coulomb as a base unit.

QUESTION A2.1

This was well answered, with around 9 in 10 students scoring the mark.

QUESTION A2.2

This was well answered, with around 7 in 10 students scoring the mark. Some students quoted the wavelength instead of four times the amplitude.

Copyright © 2026 Oxford International AQA Examinations and its licensors. All rights reserved.

QUESTION A3.1

This was well answered, with around 8 in 10 students scoring the mark. A description of one data point, for example, ‘when temperature is high, resistance is low’ did not score the mark.

QUESTION A3.2

This was well answered, with around 8 in 10 students scoring at least two of the three marks. Many students opted to sketch a graph. The idea of a sharp increase in resistance at the critical temperature was the mark many students missed.

QUESTION A4

Nearly all students attempted this question and around 6 in 10 scored full marks. A common misconception seen was that the diffraction orders are equally spaced by angle, with students simply tripling the first order angle to obtain the third order angle. A small minority of students attempted to use the double slit equation, which is not valid for a diffraction grating.

QUESTION A5.1

Most students scored at least 1 mark in this question, but only around 4 in 10 students scored both. A minority of students did not attempt to state the condition that temperature remains constant. Some students simply stated $V = IR$ which defines resistance and is not enough to imply voltage is proportional to current without R being stated as constant.

QUESTION A5.2

The majority of students scored at least two marks in this question, but only around half of students scored all three. Common errors included sketching the curve incorrectly, particularly with a flat gradient at 12V.

QUESTION A6.1

This was well answered, with around 8 in 10 students scoring both marks. A small proportion converted the unit incorrectly.

QUESTION A6.2

Around 7 in 10 students scored full marks in this calculation. One common error was using equations that apply to photons (e.g. $E = \frac{hc}{\lambda}$) rather than the de Broglie wavelength for electrons.

Copyright © 2026 Oxford International AQA Examinations and its licensors. All rights reserved.

QUESTION A6.3

This question was the first in the paper that most students found harder. Around two thirds scored at least one mark, but only around 1 in 6 students scored all three marks. Most students identified that f_B is below the threshold frequency, but fewer related this to photon energy or commented on the one to one interaction.

QUESTION A7.1

Around 3 in 5 students scored this mark; some lost the mark due to a significant figure penalty. In general, students may be penalised if they quote a numerical answer to just one significant figure, where data is given to more significant figures.

QUESTION A7.2

Around 3 in 5 students scored both marks in this calculation. The common mistake was calculating the extension due to m with no further calculation.

QUESTION A7.3

This was well answered, with nearly 8 in 10 students scoring full marks. A small proportion of students confused time period and frequency.

QUESTION A7.4

This question was well answered, with most students scoring at least 2 of the 3 marks. Most students were able to identify how the amplitude and frequency would change, but some did not give a full enough explanation to score the third mark.

QUESTION A8.1

As in previous series, this definition question was not well answered, with only 3 in 10 students scoring the mark. Many answered in terms of terminal potential difference rather than emf. This was one of the lowest-scoring questions in the paper.

QUESTION A8.2

Students also found this question difficult. Around half of the students scored at least one mark; only a quarter scored both. Many students did not specify that the voltmeter must be high resistance.

Copyright © 2026 Oxford International AQA Examinations and its licensors. All rights reserved.

QUESTION A8.3

This level of response question received an even spread of responses, with approximately equal numbers of students scoring each of the marks 0-6. Some of the weaker responses treated the circuit as having no internal resistance. Some responses stated the changes to **A_1**, **A_2** and **V** without any further explanation.

A common misconception was that the cell outputs a set current that 'splits' between the branches; therefore when a second branch is added, the current in the original branch decreases but the total stays the same. This is a misconception also seen in previous series.

QUESTION A9.1

Around half the students scored the mark for this question. Many suggested material dispersion, which does not apply to monochromatic light.

QUESTION A9.2

As in previous series, the algebraic derivation was well answered, with around two thirds of students scoring full marks. Most knew which equations were relevant, but some manipulated them unconvincingly, for example confusing the two speeds. The best responses used written words to justify each new line of working.

QUESTION A9.3

This 3-mark calculation differentiated well between students, and around two thirds of students scored at least 2 out of 3. The common error was using c, the speed of light in a vacuum, for the speed of light in the fibre.

QUESTION A9.4

This question yielded some well-worded responses, but also identified a key misconception. Many students suggested that 'cladding keeps more light in the fibre' or enables 'more total internal reflection', which is essentially the opposite of what happens and, if true, would not reduce modal dispersion. This misconception has also been seen in previous series. Students who made this incorrect statement were limited to one marking point. Around half of students scored this point but only a quarter scored 2 or more marks. This made this one of the lowest scoring questions in the paper.

QUESTION A9.5

This question was found to be the hardest on the paper, with only 1 in 5 students scoring any marks, and only small number scoring both. Most students drew pulses with a larger amplitude, consistent with the misconception that 'cladding keeps more light in the fibre'. Many students also drew pulses with narrow peaks but unchanged base widths. Some

Copyright © 2026 Oxford International AQA Examinations and its licensors. All rights reserved.

students drew pulses with a decreased time period, perhaps confusing the idea that cladding can enable a higher frequency of transmission.

QUESTION B10.1

Around 3 in 4 students scored at least one mark, but only around a third scored both. Common errors was miscounting the number of fringe spaces. Another error was rounding to an unsuitable number of significant figures.

QUESTION B10.2

Most students (around 4 in 5) scored at least one mark in this question, but under half scored both. Again, a common error was rounding to an unsuitable number of significant figures.

QUESTION B10.3

This question was better answered with around two thirds scoring the mark. Examiners checked each calculation so that no students were penalised here for their error in 10.1.

QUESTION B10.4

Around 3 in 4 students scored at least one mark, but only around a third scored both. Again, a common error was rounding to an unsuitable number of significant figures.

QUESTION B11.1

Around a third of students scored the mark for this question. Many referred vaguely to ‘the angle’ without distinguishing between the angle on the x-axis of the graph, and the angle between the transmission axes of the polarising filters, which were not the same. Some students referred to Malus’ Law, which is not required on our specification; this was treated as neutral as the student still had to specify clearly that ‘the angle’ referred to the angle between the transmission axes of the filters.

QUESTION B11.2

Most students scored the first marking point but under half scored the second. Some stated that the current increases, which was not considered a full explanation.

Copyright © 2026 Oxford International AQA Examinations and its licensors. All rights reserved.

QUESTION B11.3

This question required combining information from two graphs and a diagram, and the cohort met this challenge very well, with three quarters scoring at least 2 out of 3 marks. The most common error was reading the resistance off the LDR graph incorrectly; the vast majority were able to apply the potential divider ratio correctly.

QUESTION B11.4

This question yielded some well worded responses that were interesting to read, and around half scored at least 2 out of 3 marks. Most students identified the issue with circuit 1. Many students had the general idea that the resistor in circuit 2 was too low a resistance, but on its own this was not enough of an explanation for a mark, without a comparison and a description of the consequence.

QUESTION C12

This was well answered, with around 4 in 5 students scoring the mark.

QUESTION C13

This was well answered, with around 7 in 10 students scoring the mark. The other options were chosen about equally.

QUESTION C14

This was well answered, with around 4 in 5 students scoring the mark.

QUESTION C15

This was well answered, with around 4 in 5 students scoring the mark..

QUESTION C16

This was well answered, with around 4 in 5 students scoring the mark.

QUESTION C17

This was well answered, with around 4 in 5 students scoring the mark..

QUESTION C18

Copyright © 2026 Oxford International AQA Examinations and its licensors. All rights reserved.

This was found to be the hardest of the multiple choice questions in this paper, with around half of the students scoring the mark. The second-most popular option was C, which corresponded with the misconception that phase varies continuously in a stationary wave.

QUESTION C19

This was fairly well answered, with around 2 in 3 students scoring the mark. The second-most popular option was D. Interestingly, when a similar question was asked several years ago, a large proportion of students had the misconception that nodes no longer form; however, option A 'antinodes no longer form' was the least popular choice here.

QUESTION C20

This was fairly well answered, with around 7 in 10 students scoring the mark. The other options were chosen about equally.

QUESTION C21

This was well answered, with around 3 in 4 students scoring the mark.

QUESTION C22

This was quite well answered, with around 7 in 10 students scoring the mark. The next most popular choices were A and C.

QUESTION C23

This was well answered, with around 7 in 10 students scoring the mark. The next most popular choice was D, in which excitation and de-excitation were confused.

QUESTION C24

This was the highest scoring multiple choice question on this paper, with over 4 in 5 students scoring the mark.

QUESTION C25

This fairly well answered but was the second-lowest scoring multiple choice question, with 3 in 5 scoring the mark. The next most popular options were A and C.

Copyright © 2026 Oxford International AQA Examinations and its licensors. All rights reserved.